

Titanic Exploratory Data Analysis (EDA)

Comprehensive Visual and Statistical Exploration

Dataset Overview and Summary

Number of Rows: 891 | Columns: 12

Basic Info:

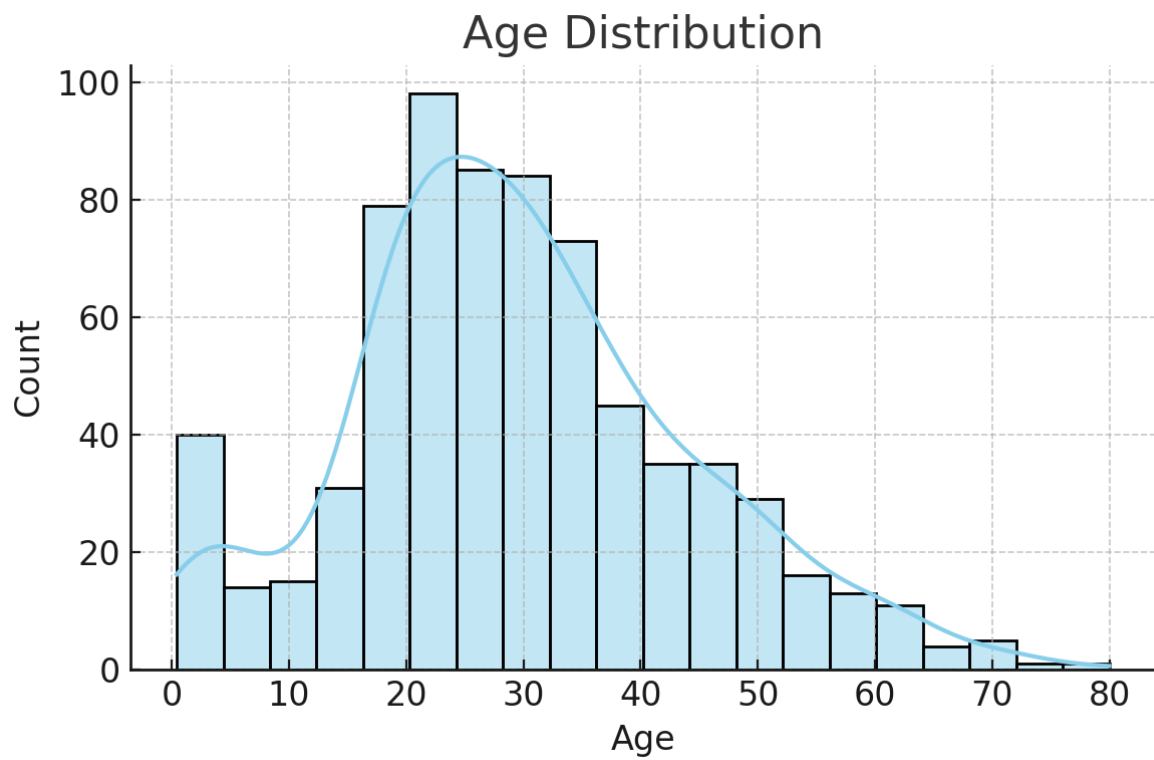
None

Missing Values Summary: PassengerId	0 Survived	0 Pclass	0 Name	0	
Sex	0 Age	177 SibSp	0 Parch	0 Ticket	0 Fare
0 Cabin	687 Embarked	2			

Descriptive Statistics

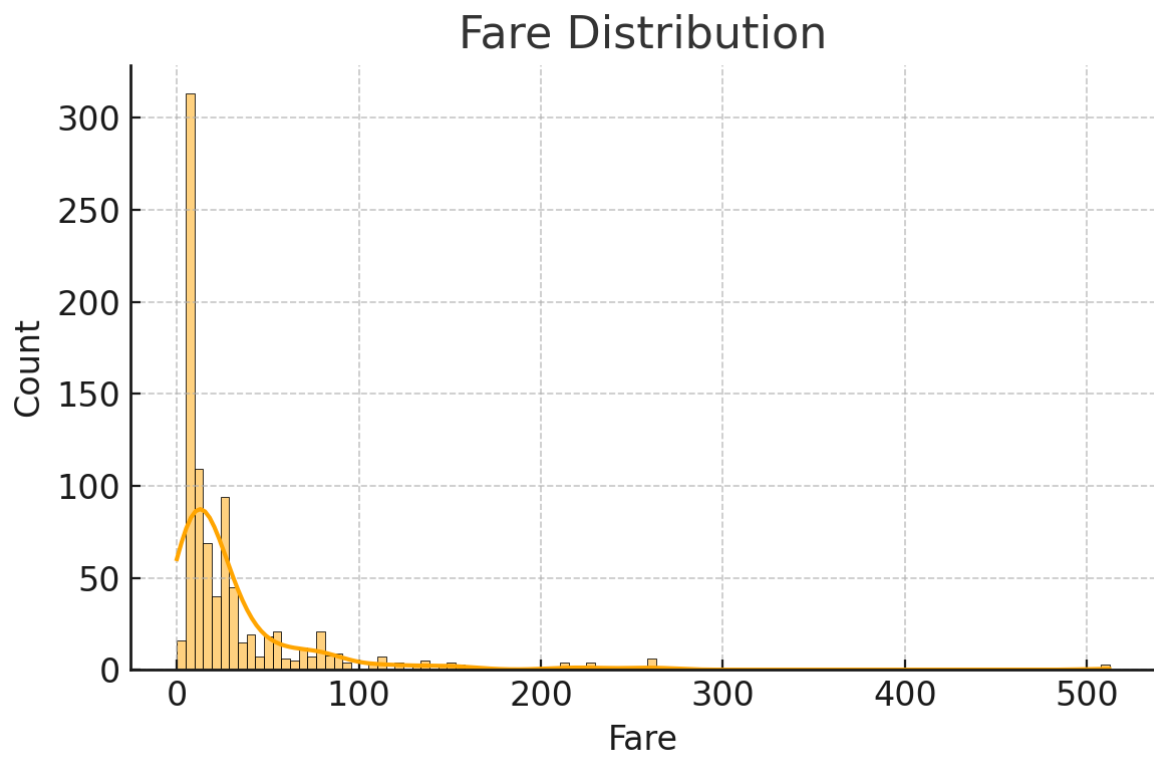
	count	unique		top	freq	mean	std	min	25%
50%	75%	max	PassengerId	891.0	NaN	NaN	NaN	446.0	257.353842
1.0	223.5	446.0	668.5	891.0	Survived	891.0	NaN	NaN	NaN
0.383838	0.486592	0.0	0.0	0.0	0.0	1.0	1.0	Pclass	891.0
NaN	NaN	2.308642	0.836071	1.0	2.0	3.0	3.0	Name	891
Braund, Mr. Owen Harris	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
891	2	male	577	NaN	NaN	NaN	NaN	NaN	NaN
NaN	Age	714.0	NaN	NaN	NaN	29.699118	14.526497	0.42	20.125
28.0	38.0	80.0	SibSp	891.0	NaN	NaN	NaN	0.523008	1.102743
0.0	0.0	0.0	1.0	8.0	Parch	891.0	NaN	NaN	NaN
0.381594	0.806057	0.0	0.0	0.0	0.0	6.0	Ticket	891	681
347082	7	NaN	NaN	NaN	NaN	NaN	NaN	Fare	891.0
NaN	NaN	32.204208	49.693429	0.0	7.9104	14.4542	31.0	512.3292	Cabin
B96	B98	4	NaN	NaN	NaN	NaN	NaN	NaN	Embarked
3		S	644	NaN	NaN	NaN	NaN	NaN	NaN

Age Distribution

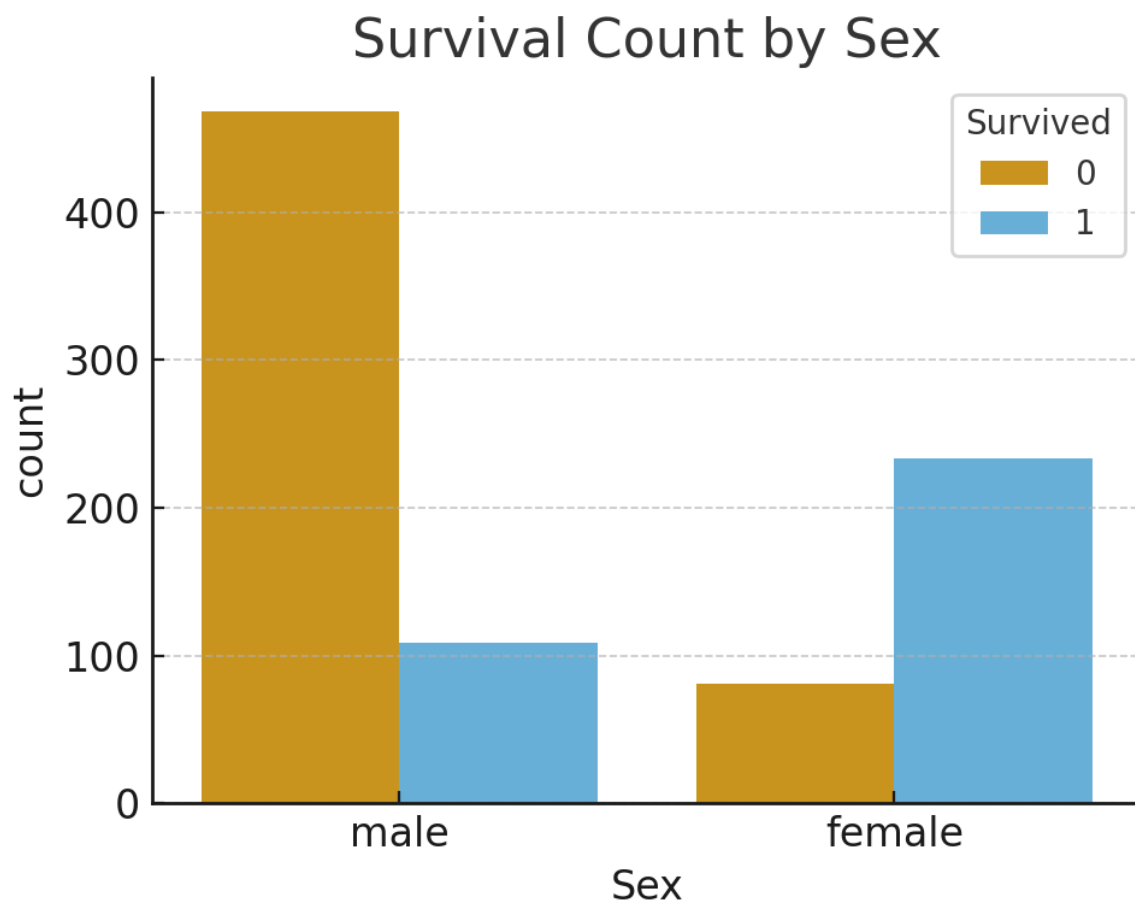


Observation: Most passengers are aged between 20-40 years. Missing ages may require imputation.

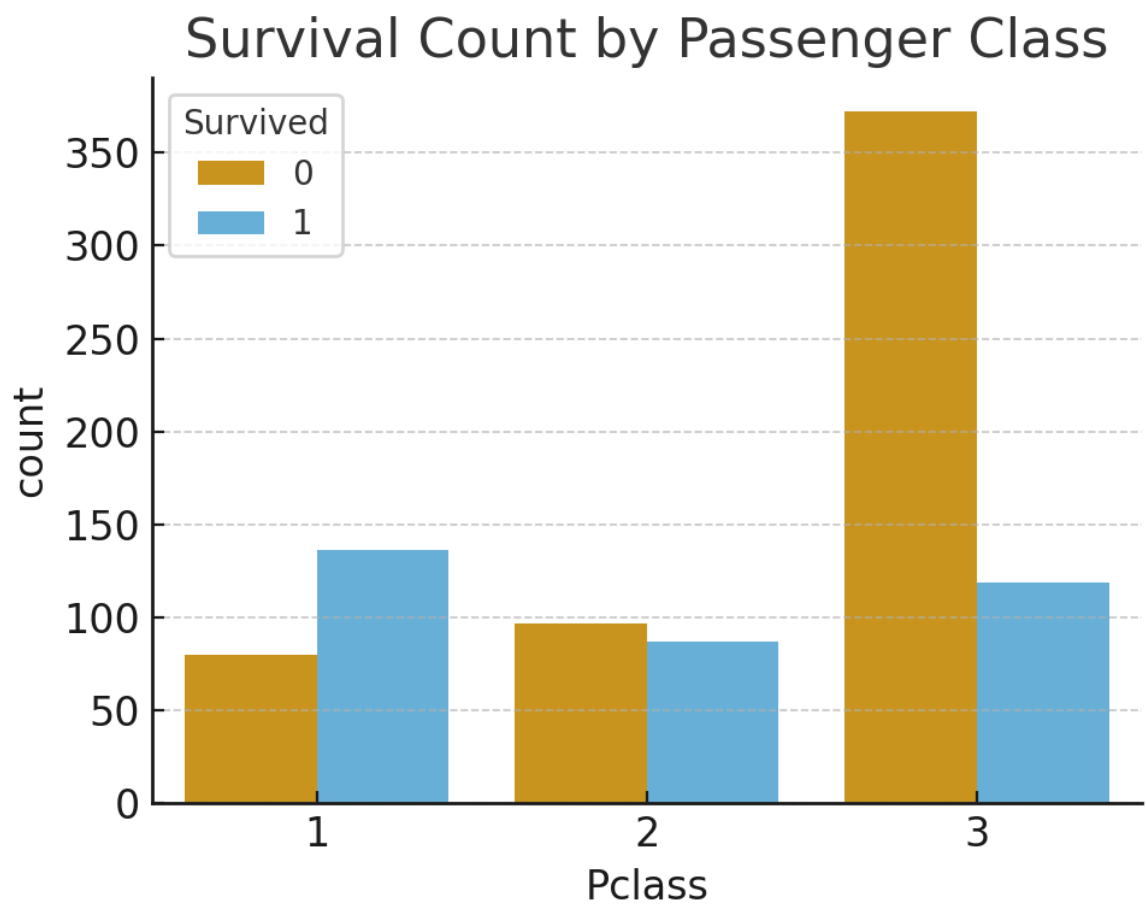
Fare Distribution



Observation: Fare distribution is right-skewed; some passengers paid significantly higher fares.

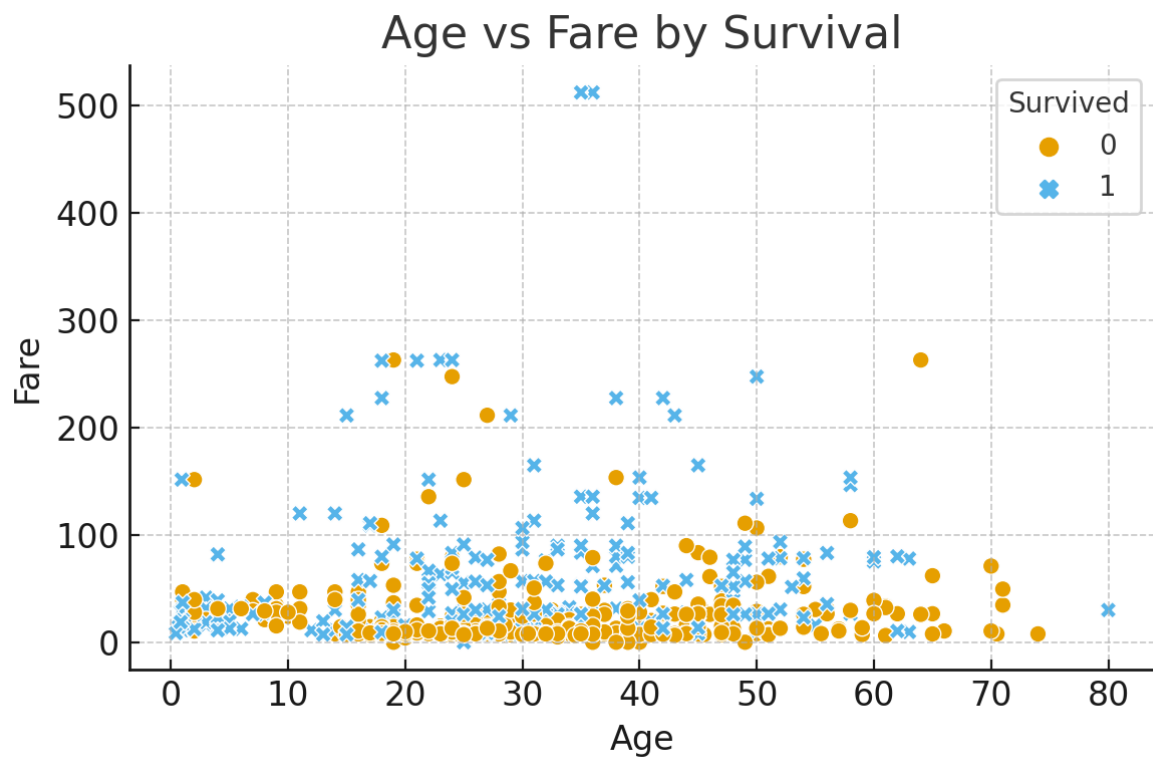


Observation: Females had a much higher survival rate compared to males.

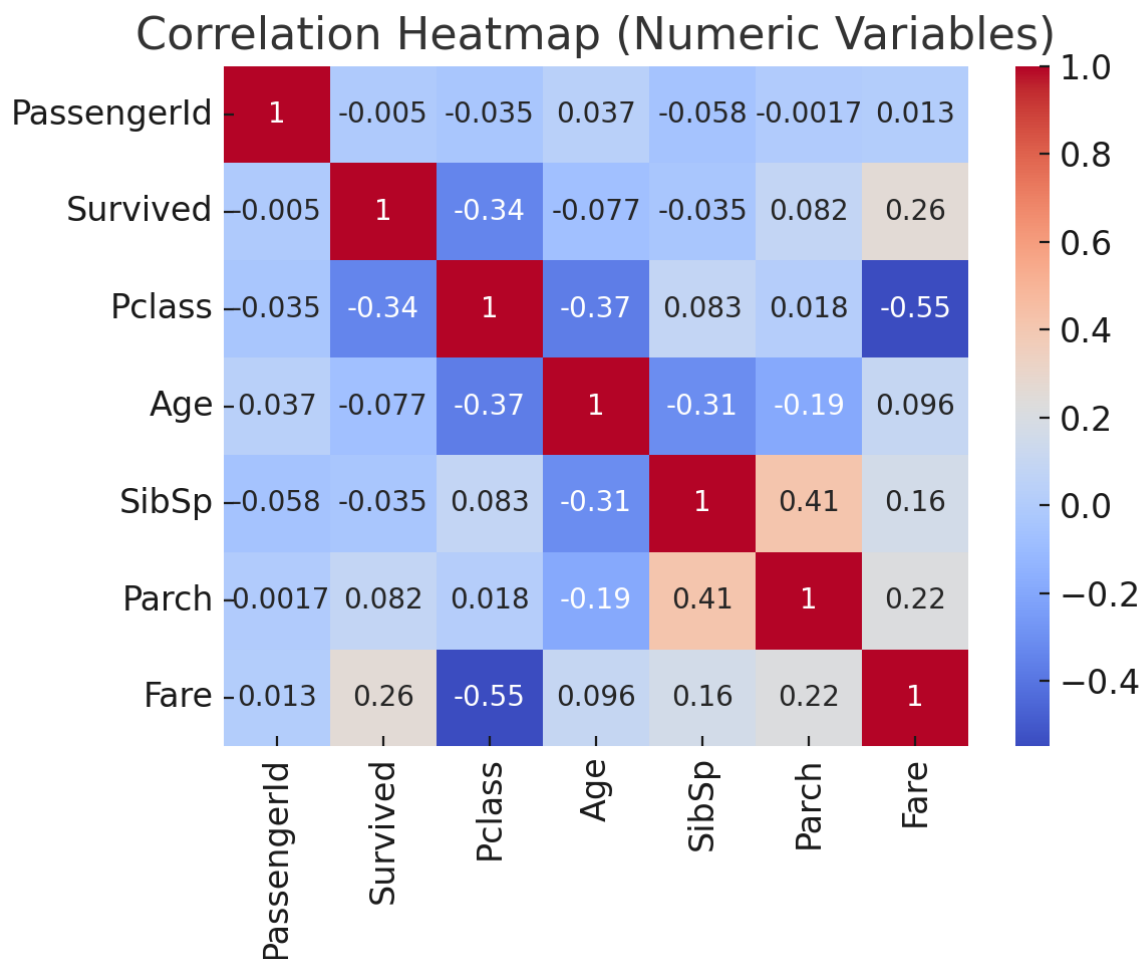


Observation: Higher-class passengers had higher survival rates.

Age vs Fare Scatter



Observation: No clear linear relation, but survivors tend to have higher fares and younger age clusters.



Observation: Fare and Passenger Class are strongly correlated with Survival.

Summary of Findings

1. Passenger Class and Fare strongly influence survival; higher class and fare passengers had better chances. 2. Females had much higher survival rates compared to males. 3. Age has missing values and a non-uniform distribution centered around 30. 4. Fare is right-skewed; log transformation may help for modeling. 5. Missing values exist in Age, Cabin, and Embarked columns and should be addressed before modeling. 6. Correlation heatmap confirms relationships among Pclass, Fare, and Survival. Further steps: feature engineering (e.g., family size, title extraction), data imputation, and model building (logistic regression, random forest).